

Lab 11 - Deployment & CI/CD

1. Introduction

Hello World

This screenshot shows the GitHub Actions interface for a workflow named 'Hello World workflow #1'. The workflow is in a 'Success' state, having been manually triggered 1 hour ago. The total duration of the workflow is 14 seconds. The interface includes a sidebar with navigation options: Summary, All jobs, Run details, Usage, and Workflow file. The main content area displays the workflow's status and a table of jobs. A job named 'hello-world-job' is listed with a duration of 2 seconds. The workflow file 'hello_world.yaml' is shown, indicating it was triggered by a 'workflow_dispatch' event.

Manually triggered 1 hour ago	Status	Total duration	Artifacts
bartoszkordek 8fd0984 main	Success	14s	-

hello_world.yaml
on: workflow_dispatch

Job	Duration
hello-world-job	2s

This screenshot shows the detailed view of the 'hello-world-job' from the previous workflow. The job is marked as 'succeeded now in 2s'. The interface includes a sidebar with navigation options: Summary, All jobs, Run details, Usage, and Workflow file. The main content area displays the job's status and a list of steps. The first step, 'Set up job', is expanded, showing a list of actions performed by the runner, including setting the runner version, provisioning the image, setting the operating system, and configuring the GitHub token. The second step, 'Print Hello World', is also expanded, showing the output of the 'echo' command. The third step, 'Complete job', is expanded, showing the cleanup of orphan processes.

hello-world-job
succeeded now in 2s

Search logs

Step	Duration
Set up job	0s
Print Hello World	0s
Complete job	0s

1 Current runner version: '2.334.0'

2 ▶ Runner Image Provisioner

9 ▶ Operating System

13 ▶ Runner Image

18 ▶ GITHUB_TOKEN Permissions

22 Secret source: Actions

23 Prepare workflow directory

24 Prepare all required actions

25 Complete job name: hello-world-job

1 ▶ Run echo "Hello world!"

4 Hello world!

1 Cleaning up orphan processes

2. Project setup

Exercise 1

```
(base) bartoszkordek@MacBook-Air-Bartosz-2 MLOps_course_lab11 % uv run pytest tests/ -v
===== test session starts =====
platform darwin -- Python 3.12.13, pytest-9.0.3, pluggy-1.6.0 -- /Users/bartoszkordek/PycharmProjects/MLOps_course_lab11/.venv/bin/python3
cachedir: .pytest_cache
rootdir: /Users/bartoszkordek/PycharmProjects/MLOps_course_lab11
configfile: pyproject.toml
plugins: anyio-4.13.0
collected 4 items

tests/test_app.py::test_sentiment_request_default_labels PASSED [ 25%]
tests/test_app.py::test_sentiment_request_custom_labels PASSED [ 50%]
tests/test_app.py::test_sentiment_response_shape PASSED [ 75%]
tests/test_app.py::test_classifier_label_pick_logic PASSED [100%]

===== 4 passed in 0.35s =====

(base) bartoszkordek@MacBook-Air-Bartosz-2 MLOps_course_lab11 % aws ecr describe-repositories --repository-names sentiment-app-onnx --region us-east-1
{
  "repositories": [
    {
      "repositoryArn": "arn:aws:ecr:us-east-1:315692715572:repository/sentiment-app-onnx",
      "registryId": "315692715572",
      "repositoryName": "sentiment-app-onnx",
      "repositoryUri": "315692715572.dkr.ecr.us-east-1.amazonaws.com/sentiment-app-onnx",
      "createdAt": "2026-05-09T10:20:22.473000+02:00",
      "imageTagMutability": "MUTABLE",
      "imageScanningConfiguration": {
        "scanOnPush": false
      }
    }
  ]
}
```

3. CI pipeline

Exercise 2

The screenshot shows the GitHub Actions interface for a workflow named 'CI/CD workflow #1'. The workflow is in a 'Success' state, having been manually triggered 5 days ago. The total duration of the run is 13 seconds. The workflow file, 'ci_cd_workflow.yaml', is shown with a single job named 'checks_and_tests' that completed successfully in 10 seconds. A warning annotation is present, stating that Node.js 20 actions are deprecated and may not work as expected. The interface includes a sidebar with navigation options like Summary, All jobs, Run details, Usage, and Workflow file. The top navigation bar shows the repository name 'bartoszkordek / MLOps_course_lab11' and various action buttons.

github.com/bartoszkordek/MLOps_course_lab1/actions/runs/25597196500

On April 24 we'll start using GitHub Copilot interaction data for AI model training unless you opt out. [Review this update](#) and manage your preferences in your [GitHub account settings](#).

← CI/CD workflow

✓ CI/CD workflow #1 Re-run all jobs

Summary

All jobs

checks_and_tests

Run details

Usage

Workflow file

Manually triggered now

bartoszkordek → 7f1c0a3 main

Status

Success

Total duration

13s

Artifacts

—

ci_cd_workflow.yaml

on: workflow_dispatch

checks_and_tests 10s

Annotations

1 warning

checks_and_tests

Node.js 20 actions are deprecated. The following actions are running on Node.js 20 and may not work as expected: actions/checkout@v4. Action...

Show more

github.com/bartoszkordek/MLOps_course_lab1/actions/runs/25597196500/job/75145039784

Summary

All jobs

checks_and_tests

Run details

Usage

Workflow file

checks_and_tests

succeeded now in 10s

Search logs

Set up job 1s

Show run info 0s

Show region 0s

Run echo "Deploying to us-east-1"

Deploying to us-east-1

Checkout code repo 1s

Run actions/checkout@v4

Syncing repository: bartoszkordek/MLOps_course_lab1

Getting Git version info

Temporarily overriding HOME='/home/runner/work/_temp/bdea1836-b16b-4ca1-b20d-0210e6be07bc' before making global git config changes

Adding repository directory to the temporary git global config as a safe directory

/usr/bin/git config --global --add safe.directory /home/runner/work/MLOps_course_lab1/MLOps_course_lab1

Deleting the contents of '/home/runner/work/MLOps_course_lab1/MLOps_course_lab1'

Initializing the repository

Disabling automatic garbage collection

Setting up auth

Fetching the repository

Determining the checkout info

/usr/bin/git sparse-checkout disable

/usr/bin/git config --local --unset-all extensions.worktreeConfig

Checking out the ref

/usr/bin/git log -1 --format=%H

7f1c0a36e956585836ef652803cfae7e634a1064

Setup uv 1s

Install dependencies

github.com/bartoszkordek/MLOps_course_lab11/actions/runs/25597196500/job/75145039784#step:5:27

Potwierdź swoją tożsamość

Summary

All jobs

checks_and_tests

Run details

Usage

Workflow file

checks_and_tests

succeeded 1 minute ago in 10s

Search logs

Checkout code repo1s

Setup uv1s

```
1 ▶ Run astral-sh/setup-uv@v7
26 Trying to find version for uv in: /home/runner/work/MLOps_course_lab11/MLOps_course_lab11/uv.toml
27 Could not find file: /home/runner/work/MLOps_course_lab11/MLOps_course_lab11/uv.toml
28 Trying to find version for uv in: /home/runner/work/MLOps_course_lab11/MLOps_course_lab11/pyproject.toml
29 Could not determine uv version from uv.toml or pyproject.toml. Falling back to latest.
30 Fetching version data from https://raw.githubusercontent.com/astral-sh/uv/main/v1/uv.ndjson ...
31 Downloading uv from "https://releases.astral.sh/github/uv/releases/download/0.11.12/uv-x86_64-unknown-linux-gnu.tar.gz" ...
32 /usr/bin/tar xz --warning=no-unknown-keyword --overwrite -C /home/runner/work/_temp/4f69db41-547c-4d73-b7eb-9c478a8698e8 -f
/home/runner/work/_temp/fd8d7421-f5dd-49c4-9fe7-fcb5342fd3e3
33 Added /home/runner/.local/bin to the path
34 Added /opt/hostedtoolcache/uv/0.11.12/x86_64 to the path
35 Set UV_PYTHON_INSTALL_DIR to /home/runner/work/_temp/uv-python-dir
36 Added /home/runner/work/_temp/uv-python-dir to the path
37 Set UV_CACHE_DIR to /home/runner/work/_temp/setup-uv-cache
38 Successfully installed uv version 0.11.12
39 Searching files using cache dependency glob:
/home/runner/work/MLOps_course_lab11/MLOps_course_lab11/**/*requirements*.txt,/home/runner/work/MLOps_course_lab11/MLOps_course_lab11/**
/*requirements*.in,/home/runner/work/MLOps_course_lab11/MLOps_course_lab11/**/*constraints*.txt,/home/runner/work/MLOps_course_lab11/MLO
ps_course_lab11/**/*constraints*.in,/home/runner/work/MLOps_course_lab11/MLOps_course_lab11/**/*pyproject.toml,/home/runner/work/MLOps_co
urse_lab11/MLOps_course_lab11/**/*uv.lock,/home/runner/work/MLOps_course_lab11/MLOps_course_lab11/**/*py.lock
40 /home/runner/work/MLOps_course_lab11/MLOps_course_lab11/pyproject.toml
41 /home/runner/work/MLOps_course_lab11/MLOps_course_lab11/uv.lock
42 Found 2 files to hash.
43 Trying to restore cache from GitHub Actions cache with key: setup-uv-2-x86_64-unknown-linux-gnu-ubuntu-24.04-3.12.3-pruned-
5ae39d8df94147384f4dafdf8d9e78fb7afbe7bb0b287e61cb61f59d777f2b57
44 No GitHub Actions cache found for key: setup-uv-2-x86_64-unknown-linux-gnu-ubuntu-24.04-3.12.3-pruned-
5ae39d8df94147384f4dafdf8d9e78fb7afbe7bb0b287e61cb61f59d777f2b57
```

github.com/bartoszkordek/MLOps_course_lab11/actions/runs/25597196500/job/75145039784#step:5:27

Potwierdź swoją tożsamość

Summary

All jobs

checks_and_tests

Run details

Usage

Workflow file

checks_and_tests

succeeded 1 minute ago in 10s

Search logs

Install dependencies1s

```
1 ▶ Run uv sync --group integration
9 Using CPython 3.12.3 interpreter at: /usr/bin/python3
10 Creating virtual environment at: .venv
11 Resolved 84 packages in 0.67ms
12 Building lab11-sentiment @ file:///home/runner/work/MLOps_course_lab11/MLOps_course_lab11
13 Downloading hf-xet (4.3MiB)
14 Downloading ruff (10.8MiB)
15 Downloading numpy (15.9MiB)
16 Downloading pygments (1.2MiB)
17 Downloading tokenizers (3.1MiB)
18 Downloading pydantic-core (2.0MiB)
19 Downloading onnxruntime (17.3MiB)
20 Downloaded pydantic-core
21 Downloaded tokenizers
22 Downloaded hf-xet
23 Downloaded ruff
24 Downloaded pygments
25 Downloaded onnxruntime
26 Downloaded numpy
27 Built lab11-sentiment @ file:///home/runner/work/MLOps_course_lab11/MLOps_course_lab11
28 Prepared 45 packages in 934ms
29 Installed 45 packages in 47ms
30 + annotated-doc==0.0.4
31 + annotated-types==0.7.0
32 + anyio==4.13.0
33 + aws-lambda-rpc-interface==4.0.0
34 + certifi==2026.4.22
35 + click==8.3.3
36 + fastapi==0.136.1
37 + filelock==3.29.0
```


Summary

All jobs

checks_and_tests

Run details

Usage

Workflow file

checks_and_tests

succeeded 1 minute ago in 10s

Search logs

Run linter (Ruff)0s

1 ▶ Run uv run ruff check .
9 All checks passed!

Run tests (Pytest)2s

1 ▶ Run uv run pytest tests/ -v
9 ===== test session starts =====
10 platform linux -- Python 3.12.3, pytest-9.0.3, pluggy-1.6.0 -- /home/runner/work/ML0ps_course_lab11/ML0ps_course_lab11/.venv/bin/python
11 cachedir: .pytest_cache
12 rootdir: /home/runner/work/ML0ps_course_lab11/ML0ps_course_lab11
13 configfile: pyproject.toml
14 plugins: anyio-4.13.0
15 collecting ... collected 4 items
16
17 tests/test_app.py::test_sentiment_request_default_labels PASSED [25%]
18 tests/test_app.py::test_sentiment_request_custom_labels PASSED [50%]
19 tests/test_app.py::test_sentiment_response_shape PASSED [75%]
20 tests/test_app.py::test_classifier_label_pick_logic PASSED [100%]
21
22 ===== 4 passed in 0.86s =====

Summary

All jobs

checks_and_tests

Run details

Usage

Workflow file

checks_and_tests

succeeded 1 minute ago in 10s

Search logs

Run tests (Pytest)2s

22 ===== 4 passed in 0.86s =====

Post Setup uv1s

1 Post job cleanup.
2 UV_CACHE_DIR is already set to /home/runner/work/_temp/setup-uv-cache
3 UV_PYTHON_INSTALL_DIR is already set to /home/runner/work/_temp/uv-python-dir
4 Pruning cache...
5 /opt/hostedtoolcache/uv/0.11.12/x86_64/uv cache prune --ci --force
6 Pruning cache at: /home/runner/work/_temp/setup-uv-cache
7 Removed 3217 files (178.8MiB)
8 Including uv cache path: /home/runner/work/_temp/setup-uv-cache
9 /usr/bin/tar --posix -cf cache.tzst --exclude cache.tzst -P -C /home/runner/work/ML0ps_course_lab11/ML0ps_course_lab11 --files-from manifest.txt --use-compress-program zstdmt
10 Sent 75877 of 75877 (100.0%), 0.4 MBs/sec
11 uv cache saved with key: setup-uv-2-x86_64-unknown-linux-gnu-ubuntu-24.04-3.12.3-pruned-5ae39d8df94147384f4dafdf8d9e78fb7afbe7bb0b287e61cb61f59d777f2b57

Post Checkout code repo0s

Complete job0s

1 Cleaning up orphan processes
2 **Warning:** Node.js 20 actions are deprecated. The following actions are running on Node.js 20 and may not work as expected: actions/checkout@v4. Actions will be forced to run with Node.js 24 by default starting June 2nd, 2026. Node.js 20 will be removed from the runner on September 16th, 2026. Please check if updated versions of these actions are available that support Node.js 24. To opt into Node.js 24 now, set the FORCE_JAVASCRIPT_ACTIONS_TO_NODE24=true environment variable on the runner or in your workflow file. Once Node.js 24 becomes the default, you can temporarily opt out by setting ACTIONS_ALLOW_USE_UNSECURE_NODE_VERSION=true. For more information see: <https://github.blog/changelog/2025-09-19-deprecation-of-node-20-on-github-actions-runners/>

4. Optimization & containerization

Exercise 3

The screenshot shows a GitHub Actions workflow run for the repository 'bartoszkordek / ML0ps_course_lab11'. The workflow is named 'CI/CD workflow #2' and is currently in a 'Success' state. It was manually triggered 2 minutes ago. The total duration of the run is 2 minutes and 1 second. The workflow consists of two jobs: 'checks_and_tests' (7s) and 'deploy_to_lambda' (1m 35s). The 'deploy_to_lambda' job is currently in progress. The workflow file is named 'ci_cd_workflow.yaml' and is located in the 'on: workflow_dispatch' event. The interface includes a sidebar with 'Summary', 'All jobs', 'Run details', 'Usage', and 'Workflow file'. The main content area shows the workflow status and a diagram of the job sequence.

```
(base) bartoszkordek@MacBook-Air-Bartosz-2 ML0ps_course_lab11 % aws ecr describe-images --repository-name sentiment-app-onnx --region us-east-1
{
  "imageDetails": [
    {
      "registryId": "315692715572",
      "repositoryName": "sentiment-app-onnx",
      "imageDigest": "sha256:e89d6682d02373314fa9fa3399078bc1d14d4da32bf440780623a85245349dfff",
      "imageTags": [
        "22389ad0e15147b626ee94de1694f6fccb966c98"
      ],
      "imageSizeInBytes": 779957993,
      "imagePushedAt": "2026-05-09T11:31:28.948000+02:00"
    }
  ]
}
...skipping...
{
  "imageDetails": [
    {
      "registryId": "315692715572",
      "repositoryName": "sentiment-app-onnx",
      "imageDigest": "sha256:e89d6682d02373314fa9fa3399078bc1d14d4da32bf440780623a85245349dfff",
      "imageTags": [
        "22389ad0e15147b626ee94de1694f6fccb966c98"
      ],
      "imageSizeInBytes": 779957993,
      "imagePushedAt": "2026-05-09T11:31:28.948000+02:00"
    }
  ]
}
```

5. Continuous deployment (AWS)

Exercise 4

← → ↻ github.com/bartoszkordek/MLOps_course_lab11/actions/runs/25598183999

Potwierdź swoją tożsamość

Summary

All jobs

checks_and_tests

deploy_to_lambda

Run details

Usage

Workflow file

Manually triggered 6 minutes ago

Status

Total duration

Artifacts

Success

2m 33s

-

ci_cd_workflow.yaml

on: workflow_dispatch

checks_and_tests 11s

deploy_to_lambda 2m 17s

Annotations

2 warnings

checks_and_tests

Node.js 20 actions are deprecated. The following actions are running on Node.js 20 and may not work as expected: actions/checkout@v4. Action...

deploy_to_lambda

Node.js 20 actions are deprecated. The following actions are running on Node.js 20 and may not work as expected: actions/checkout@v4, aws-ac...

← → ↻ github.com/bartoszkordek/MLOps_course_lab11/actions/runs/25598183999/job/75147629881

Potwierdź swoją tożsamość

← CI/CD workflow

CI/CD workflow #3

Re-run all jobs

Summary

All jobs

checks_and_tests

deploy_to_lambda

Run details

Usage

Workflow file

Annotations

1 warning

checks_and_tests

succeeded 9 minutes ago in 11s

Set up job 2s

Show run info 0s

Show region 0s

Checkout code repo 0s

Setup uv 2s

Install dependencies 2s

Run linter (Ruff) 0s

Run tests (Pytest) 2s

Post Setup uv 0s

Post Checkout code repo 0s

Complete job 0s

← CI/CD workflow

✓ CI/CD workflow #3

Re-run all jobs ⋮

Summary

All jobs

✓ checks_and_tests

✓ **deploy_to_lambda**

Run details

Usage

Workflow file

Annotations

1 warning

deploy_to_lambda

succeeded 6 minutes ago in 2m 17s

🔍 Search logs



> ✓ Set up job	1s
> ✓ Checkout code repo	1s
> ✓ Configure AWS credentials	0s
> ✓ Setup uv	1s
> ✓ Install deployment dependencies	19s
> ✓ Prepare model from HuggingFace Hub	14s
> ✓ Export model to ONNX	9s
> ✓ Login to ECR	1s
> ✓ Build and Push Docker image	39s
> ✓ Delete stack if in rollback state	4s
> ✓ Resolve Lambda role ARN from name	1s



Summary

All jobs

✓ checks_and_tests

✓ **deploy_to_lambda**

Run details

Usage

Workflow file

deploy_to_lambda

succeeded 6 minutes ago in 2m 17s

🔍 Search logs



> ✓ Checkout code repo	1s
> ✓ Configure AWS credentials	0s
> ✓ Setup uv	1s
> ✓ Install deployment dependencies	19s
> ✓ Prepare model from HuggingFace Hub	14s
> ✓ Export model to ONNX	9s
> ✓ Login to ECR	1s
> ✓ Build and Push Docker image	39s
> ✓ Delete stack if in rollback state	4s
> ✓ Resolve Lambda role ARN from name	1s
> ✓ Deploy with AWS SAM	44s
> ✓ Post Login to ECR	0s
> ✓ Post Setup uv	0s
> ✓ Post Configure AWS credentials	0s
> ✓ Post Checkout code repo	0s
> ✓ Complete job	0s



Summary

All jobs

checks_and_tests

deploy_to_lambda

Run details

Usage

Workflow file

deploy_to_lambda

succeeded 11 minutes ago in 2m 17s

Delete stack if in rolli

Setup uv1s

Install deployment dependencies19s

Prepare model from HuggingFace Hub14s

Export model to ONNX9s

Login to ECR1s

Build and Push Docker image39s

Delete stack if in rollback state4s

Resolve Lambda role ARN from name1s

```
1  Run bash scripts/recover_rollback.sh
2  bash scripts/recover_rollback.sh
3  shell: /usr/bin/bash -e {0}
4  env:
5    AWS_REGION: us-east-1
6    ECR_REPOSITORY: sentiment-app-onnx
7    AWS_DEFAULT_REGION: us-east-1
8    AWS_ACCESS_KEY_ID: ***
9    AWS_SECRET_ACCESS_KEY: ***
10   AWS_SESSION_TOKEN: ***
11   UV_PYTHON_INSTALL_DIR: /home/runner/work/_temp/uv-python-dir
12   UV_CACHE_DIR: /home/runner/work/_temp/setup-uv-cache
13   STACK_NAME: sentiment-app-stack
14   Current stack status: NOT_FOUND
```

Summary

All jobs

checks_and_tests

deploy_to_lambda

Run details

Usage

Workflow file

deploy_to_lambda

succeeded 3 minutes ago in 2m 17s

Search logs

Deploy with AWS SAM44s

91 CREATE_COMPLETE AWS::ApiGatewayV2::Sta ServerlessHttpApiApiGa -
92 ge tewayDefaultStage
93 CREATE_COMPLETE AWS::CloudFormation::S sentiment-app-stack -
94 tack
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97 CloudFormation outputs from deployed stack
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99 Outputs
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101 Key ApiUrl
102 Description -
103 Value <https://us8u1h4740.execute-api.us-east-1.amazonaws.com/predict>
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107 Successfully created/updated stack - sentiment-app-stack in us-east-1
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109
110 SAM CLI update available (1.159.1); (1.158.0 installed)
111 To download: <https://docs.aws.amazon.com/serverless-application-model/latest/developerguide/serverless-sam-cli-install.html>
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us-east-1.console.aws.amazon.com/ecr/repositories/private/315692715572/sentiment-app-onnx/_details?region=us-east-1

Potwierdź swoją tożsamość

Amazon ECR > Private registry > Repositories > sentiment-app-onnx

Amazon Elastic Container Registry

Private registry

Repositories

sentiment-app-onnx

Features & Settings

Public registry

Repositories

Settings

ECR public gallery [L](#)

Amazon ECS [L](#)

Amazon EKS [L](#)

Getting started [L](#)

sentiment-app-onnx

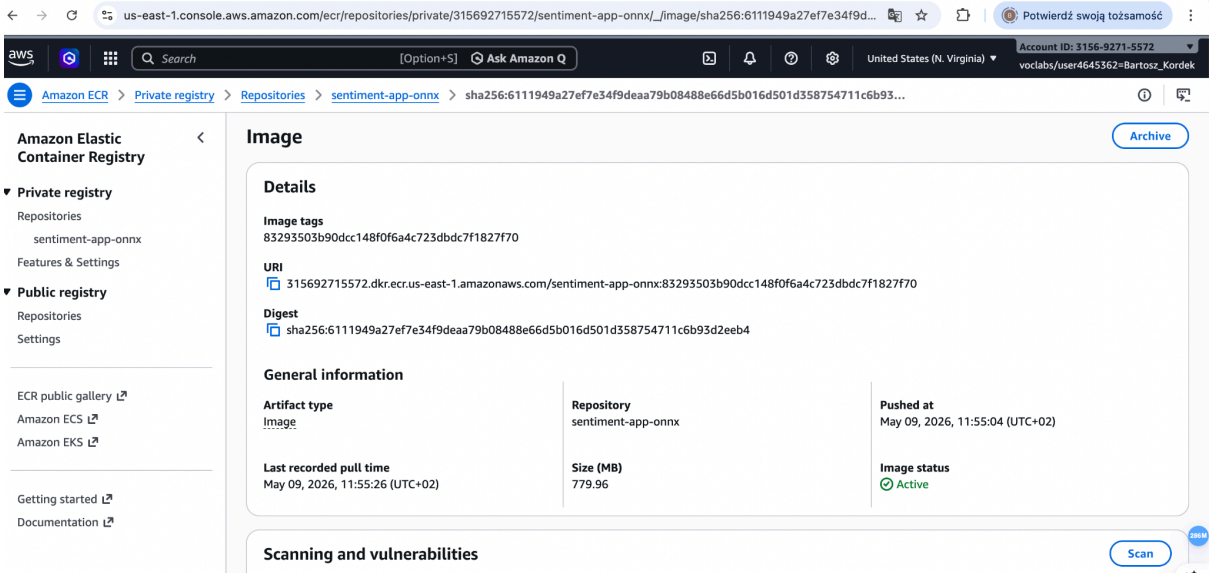
Summary Images Lifecycle policy Permissions Repository tags

Images (2) Info

Filter active images

	Image tags	Type	Created at	Image size (MB)	Image digest	Last pulled at
	83293503b90dc148f0f6a4c723dbdc7f1827f70	Image	May 09, 2026, 11:55:04 (UTC+02)	779.96	sha256:61119...	May 09, 2026, 11:55:26 (UTC+02)
	22389ad0e15147b626ee94de1694f6fccb966c98	Image	May 09, 2026, 11:31:28 (UTC+02)	779.96	sha256:e89d6...	-


```
(base) bartoszkordek@MacBook-Air-Bartosz-2 ~ % curl -X POST https://us9ujh4740.execute-api.us-east-1.amazonaws.com/predict \
  -H "Content-Type: application/json" \
  -d '{"text": "MLOps is amazing!"}'
{"label": "positive"}
(base) bartoszkordek@MacBook-Air-Bartosz-2 ~ %
```



```
(base) bartoszkordek@MacBook-Air-Bartosz-2 ~ % aws cloudformation delete-stack --stack-name sentiment-app-stack --region us-east-1
(base) bartoszkordek@MacBook-Air-Bartosz-2 ~ % aws cloudformation wait stack-delete-complete --stack-name sentiment-app-stack --region us-east-1
```

```
(base) bartoszkordek@MacBook-Air-Bartosz-2 ~ % aws ecr delete-repository --repository-name sentiment-app-onnx --region us-east-1 --force
```

```
{
  "repository": {
    "repositoryArn": "arn:aws:ecr:us-east-1:315692715572:repository/sentiment-app-onnx",
    "registryId": "315692715572",
    "repositoryName": "sentiment-app-onnx",
    "repositoryUri": "315692715572.dkr.ecr.us-east-1.amazonaws.com/sentiment-app-onnx",
    "createdAt": "2026-05-09T10:20:22.473000+02:00",
    "imageTagMutability": "MUTABLE"
  }
}
```

```
(base) bartoszkordek@MacBook-Air-Bartosz-2 ~ %
```